

Sudipta Paul

Ph.D. Candidate, Materials Science & Engineering, University of Wisconsin–Madison

Density Functional Theory | Machine-Learning Interatomic Potentials | Atomistic Simulation | Scientific Computing

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PROFESSIONAL SUMMARY

Ph.D. candidate in Materials Science and Engineering specializing in machine-learning interatomic potentials, density functional theory (DFT), *ab initio* molecular dynamics, and large-scale atomistic simulation for molten salt and high entropy ceramic systems. Experienced in developing and applying machine-learning models for scientific data analysis, property prediction, and physics-informed materials modeling using Python-based workflows. Proficient in high-performance computing and computational tools including LAMMPS, VASP, ASE, OVITO, PyTorch, and scikit-learn.

TECHNICAL SKILLS

Programming: Python, Bash

Atomistic Simulation Tools: LAMMPS, VASP, ASE, OVITO, SRIM

Engineering Software: Tecplot 360, SolidWorks, MS Office

Machine Learning: PyTorch, TensorFlow, scikit-learn

Data Analysis: Pandas, NumPy, Matplotlib

HPC Systems: Slurm, PBS

EDUCATION

Ph.D. in Materials Science and Engineering

University of Wisconsin-Madison, Madison, WI, USA

Advisor: Prof. Izabela Szlufarska

Jan 2023 – Dec 2027

GPA: 3.875/4

M.S. in Mechanical Engineering

Bangladesh University of Engineering & Technology, Bangladesh

Advisor: Prof. Mohammad Nasim Hasan

Jun 2019 – Mar 2022

GPA: 3.92/4

B.S. in Mechanical Engineering

Bangladesh University of Engineering & Technology, Bangladesh

Advisor: Prof. Mohammad Nasim Hasan

Jul 2014 – Oct 2018

GPA: 3.80/4

RESEARCH EXPERIENCE

- **Machine Learning Interatomic Potentials for Molten Salt System** | Jan 2023 – July 2026 | DOE NEUP grant #00011242
 - Developed atomic cluster expansion (ACE) machine-learning interatomic potential (MLIP) trained on DFT data to generate a thermophysical properties database with local structural features for the full compositional space of binary and ternary molten salt systems for molten salt reactor applications.
 - Analyzed the effect of transmutation products on thermophysical properties through semi-empirical models using experimental data and computational data from *ab initio* molecular dynamics (AIMD) and MLIP.
- **Radiation Effects in High Entropy Carbides** | Apr 2025 – Present | DOE grant #DE-FG02-08ER46493
 - Developed machine-learning interatomic potentials for (NbTiVMo)₄C and (NbTiVZr)₄C high entropy carbide systems.
 - Performed cascade simulation with primary knock-on atom (PKA) using ZBL potential augmented MLIP to analyze the defect evolution and microstructural changes under irradiation.
 - Investigated electron-phonon interaction effects on displacement cascades and predicted phase stability in high-entropy ceramics using free energy calculations from AIMD.
- **Atomistic Simulation on Thin Film Evaporation and Nanoscale Condensation** (Jan 2018 – Oct 2018) and (May 2021 – Jun 2022)
 - Evaluated how hydrophilicity, wetting contrast, and patterned or gradient topologies influence nanoscale evaporation and condensation heat transfer characteristics
 - Compared condensation behavior across hybrid wetting surfaces and identified atomistic mechanisms governing performance enhancement

SELECTED PUBLICATIONS*

- **Sudipta Paul**, Siamak Attarian, Dane Morgan, David Andersson, Izabela Szlufarska. "A molecular dynamics study on local structural features and thermophysical properties of the NaCl-UCl₃ molten salt with machine learning potentials." *Journal of Molecular Liquids* (2026): 129638
- **Sudipta Paul**, Siamak Attarian, Massimiliano Fratoni, Dane Morgan, Izabela Szlufarska. "Influence of transmutation products on the thermophysical properties of eutectic NaCl-UCl₃ fuel salt in a fast-spectrum molten salt reactor." *Journal of Nuclear Materials* (2025): 155572.
- **Sudipta Paul**, Mohammad Nasim Hasan. "Molecular dynamics perspective of condensation over a hybrid wetting surface." *Molecular Simulation* 48, no. 5 (2022): 448-461.
- **Sudipta Paul**, Mohammad Nasim Hasan. "Effects of gradient and patterned wetting configuration on condensation over hybrid wetting surface from atomistic viewpoint." *Journal of Molecular Liquids* (2022): 119347.
- **Sudipta Paul**, Debolina Chakraborty, Sumaiya Jannat Esha and Mohammad Nasim Hasan. "Role of wettability contrast on nanoscale condensation over hybrid wetting surface with gradient and patterned wetting configuration at various philic-phobic content." *Surfaces and Interfaces* 36, (2023):102417.

*Please see my [google scholar](#) page for the complete list

MANUSCRIPTS IN PREPARATION

- **Sudipta Paul**, Dane Morgan, Izabela Szlufarska. “Predicting thermophysical and structural properties of the NaCl–UCl₃–PuCl₃ molten salt system with machine-learning interatomic potentials (MLIPs)”.
- **Sudipta Paul**, Muhammad Waqas Qureshi, Izabela Szlufarska. “Atomistic investigation of primary radiation damage in random and short-range-ordered High-Entropy Carbides”.

WORK EXPERIENCE

Graduate Research Assistant, Department of Materials Science and Engineering University of Wisconsin-Madison, USA	Jan 2023 – Present
Lecturer, Department of Mechanical Engineering Hajee Mohammad Danesh Science and Technology University, Bangladesh	Mar 2020 – Jan 2023

RESEARCH INTERESTS

Thermophysical properties, Molten salts, Density functional theory (DFT), Machine learning interatomic potential, Molecular dynamics (MD) simulation, Radiation damage, Phase change heat transfer

AWARDS AND HONORS

- Dean’s Scholarship award for session 2015-16 and 2016-17 (Funded by BUET)
 - SSC Scholarship, Zilla Talent in 2011 (Funded by Govt. of Bangladesh)
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LEADERSHIP AND OUTREACH

- Member: Badhon Blood Donating Club, BUET, Satyen Bose Science Club, BUET, Bangladesh
 - Member: Bangladesh Students Association, University of Wisconsin-Madison, Madison, WI, USA
 - Reviewer: Applied Thermal Engineering (Elsevier)
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CITIZENSHIP STATUS

- **Citizenship:** Bangladesh
 - **U.S. Work Authorization:** F-1 student status (eligible for CPT or OPT)
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REFERENCES

Prof. Izabela Szlufarska
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Prof. Dane Morgan
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